

The Economics of Monetary Taste

An Empirical Theory of Revealed Currency Preference

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Abstract

This treatise develops a theoreticoanalytical framework for studying “monetary taste”: the latent preference structure through which households, firms, financial institutions, and states select among currencies and monetary assets. The central distinction is between monetary exposure and monetary preference. A country may exhibit high exposure to a currency because of contractual, institutional, financial, or geopolitical constraints without possessing an equivalent underlying preference for that currency. Conversely, agents may voluntarily hold a currency because it provides liquidity, stability, convertibility, collateral value, network access, or institutional credibility.

The framework combines revealed preference, monetary economics, international finance, network economics, institutional economics, behavioural science, econometrics, statistics, computer science, machine learning, evolutionary reasoning, and strategic analysis. Monetary taste is represented as a latent multidimensional object and is inferred from observed portfolio, invoicing, reserve, settlement, borrowing, pricing, and contractual choices. The resulting framework distinguishes preference from constraint, network effects from fundamental valuation, and monetary dependence from voluntary monetary choice. The analysis also shows why the analogy between currency dependence and addiction can be heuristically useful while remaining distinct from a clinical diagnosis.

The treatise ends with “The End”

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1 Introduction

The international monetary system is frequently described in terms of currency dominance, reserve-currency status, dollarization, internationalization, or dependence on a particular currency. These concepts are useful, but they do not necessarily identify the underlying preferences of economic agents.

The central proposition of this paper is therefore:

$$\boxed{\text{Observed monetary exposure} \neq \text{underlying monetary preference.}} \quad (1)$$

A country may use a currency because it is liquid, because its debts are denominated in it, because its trading partners invoice in it, because financial markets are organized around it, or because alternative instruments are unavailable. These mechanisms are economically distinct even when they generate similar observed data.

The appropriate empirical question is consequently not merely

$$\text{How dependent is country } i \text{ on currency } c? \quad (2)$$

but rather

$$\text{Why does country } i \text{ choose, retain, or tolerate currency } c? \quad (3)$$

This paper calls the underlying preference structure *monetary taste*.

The phrase has two meanings. In the economic sense, taste denotes a preference ordering over monetary characteristics. In the ordinary sense, it denotes the “palate” an economic system develops for particular monetary instruments.

The distinction is consequential. Network effects can make a monetary instrument widely used without making every participant intrinsically prefer it. Conversely, a monetary instrument may be preferred by agents even when institutional constraints prevent extensive use.

2 The Concept of Monetary Taste

Let $\mathcal{M}_i$ denote the set of monetary instruments accessible to economic unit i . These instruments may include domestic currency, foreign currencies, bank deposits, government securities, cryptocurrencies, commodities used as monetary assets, and other liquid claims.

Define the monetary taste vector of unit i as

$$\mathbf{T}_i = (T_{i1}, T_{i2}, \dots, T_{ik}). \quad (4)$$

The dimensions may include:

$$\mathbf{T}_i = (\text{liquidity, stability, return, convertibility, collateral,} \\ \text{transaction cost, network access, institutional trust, strategic value}). \quad (5)$$

The monetary choice of an agent is then represented as

$$m_i^* = \arg \max_{m \in \mathcal{M}_i} U_i(m \mid \mathbf{T}_i, \mathbf{C}_i), \quad (6)$$

where $\mathbf{C}_i$ represents constraints.

This distinction is fundamental:

$$\boxed{\mathbf{T}_i \neq \mathbf{C}_i.} \quad (7)$$

Preference describes what an agent would choose under comparable conditions. Constraint describes what the agent is capable of choosing.

3 From Monetary Exposure to Revealed Preference

Traditional measures of currency dependence can be written as an exposure vector

$$\mathbf{E}_i = (E_{i1}, E_{i2}, \dots, E_{in}). \quad (8)$$

Examples include:

- the share of trade invoiced in currency c ;
- the share of reserves held in currency c ;
- the share of external debt denominated in currency c ;
- the share of banking liabilities denominated in currency c ;
- the share of cross-border payments settled in currency c ;
- the share of international contracts denominated in currency c .

Exposure, however, may arise from equilibrium constraints rather than pure preference. A useful decomposition is

$$E_{ic} = P_{ic} + N_{ic} + I_{ic} + F_{ic} + G_{ic} + \varepsilon_{ic}, \quad (9)$$

where:

- P_{ic} represents preference;
- N_{ic} represents network effects;
- I_{ic} represents institutional constraints;
- F_{ic} represents financial constraints;
- G_{ic} represents geopolitical and strategic constraints;
- ε_{ic} represents unobserved factors.

The empirical task is therefore to recover P_{ic} from observations of E_{ic} and relevant explanatory variables.

This is a latent-variable problem.

4 Revealed Preference

The framework draws on revealed-preference theory: observed choices contain information about underlying preferences when the relevant choice set and constraints are sufficiently characterized [11, 13].

Suppose an agent chooses monetary asset m_a when m_b is also available under comparable conditions. A revealed-preference relation may be written

$$m_a \succeq_i m_b. \quad (10)$$

Repeated observations generate a revealed-preference graph

$$G_i = (V_i, E_i), \quad (11)$$

where monetary instruments constitute nodes and observed preference relations constitute directed edges.

A sequence

$$m_1 \succeq_i m_2 \succeq_i \dots \succeq_i m_k \quad (12)$$

contains information about the agent's ordering.

The empirical problem is complicated by the fact that monetary choices are rarely made under identical budget sets. Hence observed choice cannot automatically be interpreted as pure preference.

5 Monetary Taste as a Random-Utility System

A random-utility representation provides a practical econometric foundation:

$$U_{im} = V_{im}(\mathbf{X}_i, \mathbf{T}_i) + \varepsilon_{im}. \quad (13)$$

The probability that agent i selects monetary instrument m is

$$\Pr(m_i = m) = \Pr(U_{im} \geq U_{ij} \quad \forall j \in \mathcal{M}_i). \quad (14)$$

A multinomial-logit approximation gives

$$\Pr(m_i = m) = \frac{\exp(V_{im})}{\sum_{j \in \mathcal{M}_i} \exp(V_{ij})}. \quad (15)$$

More flexible specifications may use multinomial probit models, mixed logit models, hierarchical Bayesian models, latent-class models, or machine-learning estimators.

6 The Monetary Taste Function

Let monetary taste depend on a vector of characteristics:

$$\begin{aligned} V_{ic} = & \beta_1 L_{ic} + \beta_2 S_{ic} + \beta_3 R_{ic} + \beta_4 K_{ic} + \beta_5 N_{ic} + \beta_6 C_{ic} \\ & + \beta_7 I_{ic} + \beta_8 G_{ic} + \beta_9 H_{ic} + \eta_{ic}, \end{aligned} \quad (16)$$

where:

$$L_{ic} : \text{liquidity}, \quad (17)$$

$$S_{ic} : \text{stability}, \quad (18)$$

$$R_{ic} : \text{return}, \quad (19)$$

$$K_{ic} : \text{convertibility}, \quad (20)$$

$$N_{ic} : \text{network accessibility}, \quad (21)$$

$$C_{ic} : \text{collateral usefulness}, \quad (22)$$

$$I_{ic} : \text{institutional credibility}, \quad (23)$$

$$G_{ic} : \text{geopolitical usefulness}, \quad (24)$$

$$H_{ic} : \text{historical familiarity}. \quad (25)$$

The vector

$$\boldsymbol{\beta} = (\beta_1, \dots, \beta_9) \quad (26)$$

can be interpreted as a measurable representation of monetary taste.

7 A Fundamental Identification Problem

The central econometric difficulty is observational equivalence.

Suppose country i holds a large quantity of currency c . This may result from:

1. preference;
2. contractual necessity;
3. debt denomination;
4. trade invoicing;
5. reserve-management convention;

6. financial-market depth;
7. collateral requirements;
8. sanctions avoidance;
9. diplomatic arrangements;
10. historical inertia;
11. network effects.

Consequently,

$$E_{ic} \gg 0 \quad (27)$$

does not imply

$$P_{ic} \gg 0. \quad (28)$$

The two quantities must be separately identified.

8 The Monetary Taste–Constraint Decomposition

Let the observed monetary decision be

$$D_i = f(\mathbf{T}_i, \mathbf{C}_i, \mathbf{N}_i, \mathbf{G}_i, \varepsilon_i), \quad (29)$$

where:

- $\mathbf{T}_i$ is preference;
- $\mathbf{C}_i$ is domestic institutional constraint;
- $\mathbf{N}_i$ is network structure;
- $\mathbf{G}_i$ is geopolitical structure.

The objective is not to infer taste from D_i alone.

Instead, one seeks variation in constraints and environments sufficient to estimate

$$\frac{\partial D_i}{\partial \mathbf{T}_i} \quad (30)$$

separately from

$$\frac{\partial D_i}{\partial \mathbf{C}_i}. \quad (31)$$

Natural experiments, policy changes, exchange-rate shocks, financial liberalization, changes in payment infrastructure, and exogenous changes in trade composition can therefore be valuable identification strategies.

9 Network Economics of Currency Choice

Currencies possess network externalities. Let n_c denote the number or economic weight of users of currency c . A stylized network value function is

$$W_c = \alpha \log(1 + n_c) + \beta Q_c + \gamma L_c - \tau_c, \quad (32)$$

where Q_c denotes institutional quality, L_c liquidity, and τ_c transaction cost.

A positive network externality implies

$$\frac{\partial W_c}{\partial n_c} > 0. \quad (33)$$

This creates a feedback mechanism:

$$\text{Use} \rightarrow \text{Liquidity} \rightarrow \text{Lower Transaction Cost} \rightarrow \text{Greater Use.} \quad (34)$$

Thus a currency may remain dominant partly because it is already dominant.

10 Path Dependence

Historical monetary choices can affect present monetary choices:

$$M_t = F(M_{t-1}, X_t, \varepsilon_t). \quad (35)$$

Repeated historical choices can generate increasing returns to adoption. This is consistent with broader theories of path dependence and institutional persistence [10, 1].

The resulting equilibrium may therefore satisfy

$$M_t \approx M_{t-1} \quad (36)$$

even after the fundamental characteristics of competing monetary systems have changed.

This does not prove that persistence is inefficient. It merely shows that historical state variables may influence current monetary choice.

11 Monetary Taste and the Quantity Theory

The distinction between monetary identity and behavioral theory is important.

The familiar relation is

$$MV = PT. \quad (37)$$

As an accounting relationship, the identity connects nominal transactions to money and velocity. The behavioral interpretation is more demanding because it imposes assumptions about velocity, money supply, prices, and output.

Empirical evidence for the United States and euro area indicates that simple monetary relationships can be unstable while broader specifications incorporating financial and institutional variables can provide additional explanatory structure [3, 4].

Consequently, monetary taste should not be reduced to money demand alone.

It includes the selection of monetary assets, contractual units of account, reserve instruments, settlement media, and financial collateral.

12 Finance and Portfolio Choice

Monetary taste is naturally related to portfolio theory [8, 9].

Let the portfolio vector be

$$\mathbf{w}_i = (w_{i1}, \dots, w_{in}), \quad (38)$$

with

$$\sum_{c=1}^n w_{ic} = 1. \quad (39)$$

An investor may maximize

$$U_i = E(R_i) - \frac{\lambda_i}{2} \text{Var}(R_i) + \phi_i L_i + \psi_i N_i, \quad (40)$$

where L_i represents liquidity and N_i network utility.

Currency selection is therefore simultaneously an asset-allocation problem and a monetary-institution problem.

13 Behavioural Economics and Psychology

Economic agents need not possess perfectly stable or internally consistent preferences. Monetary choices can be influenced by heuristics, salience, framing, loss aversion, ambiguity, familiarity, and reference dependence [5, 6].

A behavioral monetary utility function may therefore be written

$$U_i = U_i^0 + B_i + F_i + A_i + H_i, \quad (41)$$

where:

- B_i represents behavioral biases;
- F_i represents framing;
- A_i represents ambiguity;
- H_i represents habit.

The framework deliberately avoids treating strong monetary attachment as a psychiatric disorder. The word “addiction” can be useful as a metaphor for a self-reinforcing dependency mechanism, but clinical addiction is a distinct medical and psychological concept.

14 Philosophy of Money

Money possesses several philosophical functions.

It may act as:

1. medium of exchange;
2. unit of account;
3. store of value;
4. social institution;
5. contractual convention;
6. political technology;
7. bearer of collective expectations.

The institutional view emphasizes that money is not simply a physical object. Its monetary character depends partly upon social recognition, legal institutions, and expectations [12].

This provides a bridge between monetary economics and philosophy: monetary value is simultaneously material, institutional, relational, and informational.

15 Physics: Monetary Systems as Networks

The language of statistical physics provides useful analogies without requiring monetary systems literally to obey physical laws.

Let p_c denote the probability of selecting currency c . An entropy-like measure is

$$H = - \sum_{c=1}^n p_c \log p_c. \quad (42)$$

Low entropy corresponds to concentration in a small number of currencies; high entropy corresponds to greater diversification.

A concentration index may be defined as

$$C = \sum_{c=1}^n p_c^2. \quad (43)$$

Thus

$$\frac{1}{n} \leq C \leq 1. \quad (44)$$

These measures describe monetary concentration, not welfare or desirability.

16 Chemistry: Monetary Compatibility

Chemical systems distinguish between intrinsic properties and interaction effects. A similar analytical distinction is useful for currencies.

Let q_c represent intrinsic monetary characteristics and q_{ij} represent interaction effects between currencies i and j .

Then a monetary system can be represented as

$$\mathcal{M} = \sum_i q_i + \sum_{i < j} q_{ij}. \quad (45)$$

The interaction term captures convertibility, hedging, cross-currency settlement, arbitrage, and portfolio substitution.

Thus currencies should not necessarily be analyzed independently.

17 Biology and Evolution

Currency networks can also be interpreted through evolutionary selection.

Let $x_c(t)$ denote the population share of currency c . A replicator equation is

$$\dot{x}_c = x_c (\pi_c - \bar{\pi}), \quad (46)$$

where π_c is the effective payoff associated with currency c .

This does not imply that currencies are biological organisms. It provides a formal analogy for systems in which successful adoption increases subsequent adoption.

Institutional inheritance may be represented as

$$T_{i,t+1} = \rho T_{i,t} + (1 - \rho) \tilde{T}_{i,t+1}, \quad (47)$$

where ρ captures persistence.

18 Political Economy

Money is also a political institution.

The state can influence monetary choice through:

- taxation;
- legal tender rules;
- capital controls;
- reserve requirements;
- financial regulation;
- payment infrastructure;
- sanctions;
- debt denomination;
- central-bank policy.

Consequently,

$$\mathbf{T}_i = f(\text{preferences, institutions, history}). \quad (48)$$

The political economy of monetary taste therefore concerns not only what agents want, but also how institutions shape the menu from which they choose.

19 Geopolitics and International Relations

International currencies can possess strategic characteristics.

For state i , define

$$G_{ic} = g(\text{trade, alliances, sanctions, security, financial access, diplomacy}). \quad (49)$$

A currency may therefore possess value as an instrument of international economic coordination independently of its domestic monetary characteristics.

International monetary architecture is consequently partly a coordination game.

For countries i and j ,

$$U_i(c_i, c_j) \quad (50)$$

may depend upon both countries' choices.

The strategic problem becomes

$$c_i^* = \arg \max_{c_i} U_i(c_i, c_{-i}). \quad (51)$$

A Nash equilibrium may emerge even when no country regards the equilibrium arrangement as uniquely optimal in isolation [7, 2].

20 Diplomacy and Monetary Pluralism

A plural international monetary system can be represented by a set

$$\mathcal{C} = \{c_1, c_2, \dots, c_n\}. \quad (52)$$

Diversification does not necessarily require the elimination of any one currency.

Instead, systemic resilience may involve

$$\text{Resilience} = f(\text{redundancy, convertibility, liquidity, institutional cooperation}). \quad (53)$$

Diplomatic coordination can reduce the costs associated with fragmentation.

21 Welfare Economics

A monetary architecture should ultimately be evaluated partly through its consequences for welfare.

Let social welfare be

$$W = \sum_i \omega_i U_i, \quad (54)$$

where ω_i represents a social weighting rule.

Monetary diversification may improve welfare when it reduces concentration risk, but diversification may also increase transaction costs.

Hence there is no general theorem that

$$\text{more currencies} \Rightarrow \text{greater welfare}. \quad (55)$$

Nor is there a general theorem that

$$\text{fewer currencies} \Rightarrow \text{greater welfare}. \quad (56)$$

The relevant comparison is between the welfare consequences of alternative monetary architectures.

22 Warfare Economics and Strategic Resilience

War introduces a different monetary objective function.

Let

$$R_i = R_i(\text{payment continuity, reserve access, import capacity, financial sanctions, domestic production}). \quad (57)$$

A state may therefore value monetary diversification for resilience rather than ordinary portfolio return.

A stylized objective is

$$\max_{\mathbf{w}} [E(R) - \lambda \text{Var}(R) + \kappa \text{Resilience}(\mathbf{w})]. \quad (58)$$

This creates a distinction between financial efficiency and strategic robustness.

23 Information Theory

Currency choice can also be interpreted as information processing.

Suppose agents observe signals X concerning future monetary conditions. Their posterior beliefs satisfy

$$p(\theta | X) = \frac{p(X | \theta)p(\theta)}{p(X)}. \quad (59)$$

Monetary taste may therefore depend upon perceived uncertainty.

The expected information value of a monetary asset can be expressed through an entropy reduction:

$$\text{IG} = H(\Theta) - H(\Theta | X). \quad (60)$$

Currencies that reduce uncertainty about settlement, liquidity, or future purchasing power may possess informational utility.

24 Computer Science Representation

A monetary ecosystem can be represented as a multilayer graph.

$$G = (V, E_T, E_F, E_P, E_D), \quad (61)$$

where:

- E_T represents trade relationships;
- E_F represents financial relationships;
- E_P represents payment relationships;
- E_D represents diplomatic or strategic relationships.

The monetary taste problem is then a graph-inference problem.

$$\mathbf{T} = \mathcal{F}(G, X, Y). \quad (62)$$

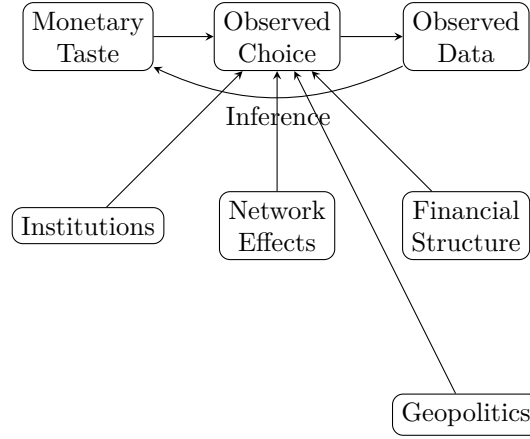


Figure 1: The empirical monetary-taste system.

25 Algorithmic Estimation

A practical estimation procedure is:

1. construct a panel of monetary choices;
2. construct measures of monetary characteristics;
3. identify institutional and geopolitical constraints;
4. estimate baseline discrete-choice models;
5. estimate latent preference factors;
6. test network effects;
7. conduct robustness analysis;
8. compare machine-learning predictions with econometric estimates;
9. distinguish predictive performance from causal identification.

Pseudocode

INPUT:

```

country-year panel
currency characteristics
trade data
reserve data
debt data
payment data
financial-market data
institutional variables
geopolitical variables

```

FOR each country i :

```

construct exposure vector  $E_i$ 
construct monetary-choice vector  $D_i$ 
construct constraint vector  $C_i$ 

```

ESTIMATE:

```

baseline discrete-choice model
latent-factor model
network-effect model

```

IDENTIFY:

preference effects
institutional effects
network effects
geopolitical effects

VALIDATE:

out-of-sample prediction
placebo tests
sensitivity analysis
alternative specifications

OUTPUT:

estimated monetary taste vector $\mathbf{T}_i$
uncertainty intervals
counterfactual monetary choices

26 Machine Learning and Latent Monetary Taste

Machine learning may be used to estimate nonlinear mappings

$$\hat{\mathbf{T}}_i = f_{\theta}(\mathbf{X}_i). \quad (63)$$

Possible methods include:

- random forests;
- gradient boosting;
- neural networks;
- autoencoders;
- graph neural networks;
- variational latent-variable models;
- Bayesian hierarchical models.

However, prediction should not be confused with identification.
A model may achieve

$$R^2 \rightarrow 1 \quad (64)$$

in a predictive environment while providing limited information about causal preference.
Thus the empirical hierarchy should be:

$$\text{Prediction} \neq \text{Identification} \neq \text{Structural Interpretation}. \quad (65)$$

27 A Latent Monetary Taste Model

Let

$$\mathbf{Z}_i = \Lambda \mathbf{T}_i + \boldsymbol{\epsilon}_i \quad (66)$$

be a factor model for observed monetary behaviors.
Here $\mathbf{Z}_i$ contains observable indicators such as:

$$\mathbf{Z}_i = (\text{reserve share, invoicing share, debt share, payment share, portfolio share}). \quad (67)$$

The latent vector $\mathbf{T}_i$ represents monetary taste.

Identification requires normalization because

$$\Lambda \mathbf{T} = (\Lambda A^{-1})(A \mathbf{T}) \quad (68)$$

for any invertible matrix A .

Consequently, factor rotations and scale normalization must be handled explicitly.

28 A Proposition on Monetary Taste

Proposition 1. Suppose monetary choice is generated by

$$D_i = f(\mathbf{T}_i, \mathbf{C}_i, \mathbf{N}_i, \mathbf{G}_i, \varepsilon_i). \quad (69)$$

Then observed monetary exposure alone does not generally identify $\mathbf{T}_i$ unless the mapping from $(\mathbf{T}_i, \mathbf{C}_i, \mathbf{N}_i, \mathbf{G}_i)$ to observed exposure is injective or additional identifying restrictions are imposed.

Proof. Suppose two distinct states

$$(\mathbf{T}_i, \mathbf{C}_i, \mathbf{N}_i, \mathbf{G}_i) \neq (\mathbf{T}'_i, \mathbf{C}'_i, \mathbf{N}'_i, \mathbf{G}'_i) \quad (70)$$

produce

$$f(\mathbf{T}_i, \mathbf{C}_i, \mathbf{N}_i, \mathbf{G}_i, \varepsilon_i) = f(\mathbf{T}'_i, \mathbf{C}'_i, \mathbf{N}'_i, \mathbf{G}'_i, \varepsilon_i). \quad (71)$$

Then the observed outcome is compatible with more than one latent preference state. Hence $\mathbf{T}_i$ cannot be uniquely recovered from exposure alone. Additional observations, restrictions, or experiments are required. $\square$

29 A Proposition on Network Persistence

Proposition 2. Suppose the utility of currency c satisfies

$$U_c = V_c + \alpha n_c, \quad \alpha > 0. \quad (72)$$

Then greater existing adoption increases the utility of subsequent adoption.

Proof. Differentiation gives

$$\frac{\partial U_c}{\partial n_c} = \alpha > 0. \quad (73)$$

Therefore existing adoption creates a positive externality for additional adoption. $\square$

The proposition does not imply that network dominance is permanent. Technological, institutional, financial, or geopolitical shocks can alter V_c and therefore change the equilibrium.

30 Empirical Measurement Architecture

A proposed empirical database is summarized in Table 1.

Table 1: Empirical dimensions of monetary taste

Dimension	Observable proxy	Possible source
Liquidity	market depth, turnover	central banks, exchanges
Stability	inflation and volatility	IMF, BIS, national statistics
Reserve taste	reserve composition	IMF COFER
Trade taste	invoicing currency	customs and trade databases
Debt taste	currency denomination	BIS, World Bank
Payment taste	settlement currency	payment-system data
Financial taste	portfolio allocation	IMF, BIS, national accounts
Network taste	counterparties	trade and banking networks
Institutional taste	credibility measures	central banks, surveys
Strategic taste	sanctions exposure	public geopolitical data

31 Identification Strategies

Several empirical strategies are possible.

31.1 Panel Estimation

A baseline panel model is

$$Y_{ict} = \alpha_i + \gamma_t + \beta_1 X_{ict} + \beta_2 N_{ct} + \beta_3 G_{ict} + \beta_4 I_{it} + \epsilon_{ict}. \quad (74)$$

Country fixed effects α_i absorb time-invariant heterogeneity, while time effects γ_t absorb common shocks.

31.2 Instrumental Variables

If monetary characteristics are endogenous,

$$X_{ict} = \pi Z_{ict} + \mathbf{W}'_{ict} \delta + u_{ict}, \quad (75)$$

followed by

$$Y_{ict} = \beta \hat{X}_{ict} + \mathbf{W}'_{ict} \gamma + \epsilon_{ict}. \quad (76)$$

The credibility of the instrument remains an empirical question.

31.3 Difference-in-Differences

When an institutional reform changes monetary choice for a treated population, a difference-in-differences design may estimate

$$Y_{it} = \alpha_i + \gamma_t + \delta(\text{Treat}_i \times \text{Post}_t) + \epsilon_{it}. \quad (77)$$

The coefficient δ identifies the treatment effect under the usual identifying assumptions.

32 Monetary Taste and Counterfactuals

Once the model is estimated, counterfactual monetary choices can be simulated.

For a change in network access,

$$\Delta N_c \neq 0, \quad (78)$$

the model predicts

$$\Delta P(c) = P(c \mid N_c + \Delta N_c) - P(c \mid N_c). \quad (79)$$

For a reduction in transaction costs,

$$\Delta \tau_c < 0, \quad (80)$$

one can estimate

$$\frac{\partial P(c)}{\partial \tau_c}. \quad (81)$$

Counterfactual analysis therefore allows monetary taste to be studied under alternative institutional architectures.

33 Monetary Taste and Systemic Risk

A monetary system can exhibit concentration risk when many agents possess correlated exposures.

Let w_{ic} denote exposure of country i to currency c . Define

$$S_c = \sum_i w_{ic}. \quad (82)$$

Systemic concentration can then be approximated by

$$SC = \sum_c S_c^2. \quad (83)$$

This is not itself a welfare measure. It is a concentration statistic.

High concentration may provide efficiency through liquidity and network effects while simultaneously increasing exposure to common shocks.

Hence the economic problem is an optimization problem involving a trade-off between

$$\text{network efficiency} \quad \text{and} \quad \text{systemic redundancy}. \quad (84)$$

34 The Addiction Metaphor

The analogy between monetary dependence and addiction should be used carefully.

A generic dependency mechanism can be represented as

$$D_t \rightarrow B_t \rightarrow R_t \rightarrow D_{t+1}, \quad (85)$$

where dependence generates behavior that reinforces subsequent dependence.

This structure occurs in many non-clinical economic systems.

The analogy becomes misleading if it is interpreted as a psychiatric diagnosis. Monetary dependence is instead a structural property of an economic network.

The analytically useful concept is therefore

$$\boxed{\text{self-reinforcing monetary dependence}} \quad (86)$$

rather than clinical addiction.

35 Monetary Taste as an Emergent Property

A particularly important conclusion follows from the preceding framework.

An individual may possess monetary preferences, but national monetary taste need not equal the arithmetic average of individual preferences.

Let individual preferences be

$$\mathbf{T}_1, \dots, \mathbf{T}_N. \quad (87)$$

The aggregate monetary taste may instead satisfy

$$\mathbf{T}^A = F(\mathbf{T}_1, \dots, \mathbf{T}_N, G, I, H), \quad (88)$$

where G is the network structure, I institutions, and H historical state.

Thus national monetary taste is potentially an emergent property.

36 A Unified Conceptual Architecture

The complete framework can be summarized as

$$\begin{aligned}
 &\text{Fundamentals} \rightarrow \text{Institutions} \rightarrow \text{Constraints}, & (89) \\
 &\text{Preferences} \rightarrow \text{Monetary Taste}, & (90) \\
 &\text{Networks} \rightarrow \text{Externalities}, & (91) \\
 &\text{Geopolitics} \rightarrow \text{Strategic Utility}, & (92) \\
 &\text{Constraints} + \text{Taste} + \text{Networks} \rightarrow \text{Observed Choice}, & (93) \\
 &\text{Observed Choice} \rightarrow \text{Data}, & (94) \\
 &\text{Data} \rightarrow \text{Econometric Inference}, & (95) \\
 &\text{Econometric Inference} \rightarrow \hat{\mathbf{T}}. & (96)
 \end{aligned}$$

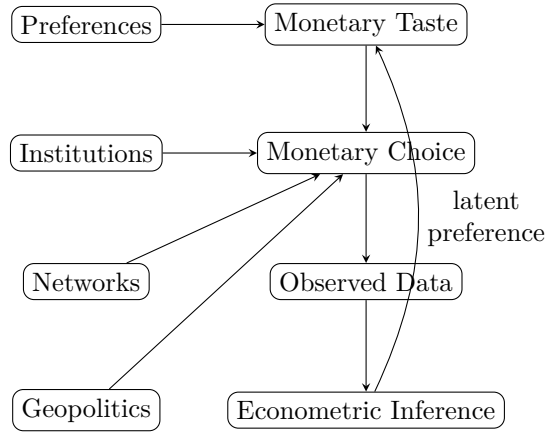


Figure 2: From preferences and institutions to revealed monetary taste.

37 Implications

The framework yields several conclusions.

First,

$$\text{currency use} \neq \text{currency preference}. \quad (97)$$

Second,

$$\text{currency dependence} \neq \text{currency preference}. \quad (98)$$

Third,

$$\text{currency dominance} \neq \text{currency superiority}. \quad (99)$$

Dominance can arise from network effects, institutional history, financial-market depth, contractual conventions, geopolitical relationships, or some combination of these mechanisms.

Fourth, monetary pluralism should not be evaluated solely by counting currencies. The relevant object is the architecture of substitution, redundancy, interoperability, liquidity, and risk sharing.

Finally, monetary taste is inherently relational. The value of a currency depends partly on what other agents accept, hold, price, settle, collateralize, and expect.

38 Research Agenda

The framework suggests a broad empirical programme.

1. Construct a global panel of currency choices.
2. Estimate country-level monetary taste vectors.
3. Separate preference from institutional constraint.
4. Estimate network externalities.
5. Identify structural breaks in monetary taste.
6. Estimate heterogeneity across households, firms, banks, and states.
7. Study monetary taste during financial crises.
8. Study monetary taste during wars and sanctions.
9. Estimate the effect of payment-system innovation.
10. Study the interaction between digital currencies and existing currency networks.
11. Construct counterfactual multi-currency equilibria.
12. Evaluate welfare and systemic-risk consequences.

A particularly useful extension would estimate

$$\mathbf{T}_{it} = F(\mathbf{T}_{i,t-1}, \mathbf{X}_{it}, \mathbf{N}_{it}, \mathbf{G}_{it}, \varepsilon_{it}), \quad (100)$$

thereby converting monetary taste from a static latent variable into a dynamic state variable.

39 Conclusion

The international monetary system should not be understood solely as a contest among currencies.

It is also a system of preferences, institutions, expectations, networks, contracts, technologies, histories, and strategic relationships.

The central empirical distinction is therefore:

$$\boxed{\text{What an economy holds} \neq \text{what an economy would choose.}} \quad (101)$$

The difference between the two is precisely where monetary taste becomes an empirical object.

A currency may be held because it is preferred. It may be held because it is required. It may be held because everyone else holds it. It may be held because financial infrastructure makes alternatives costly. It may be held because contracts were written in it decades earlier. It may be held because geopolitical circumstances make it useful.

These mechanisms produce observationally similar outcomes but imply different economic interpretations.

The task of empirical monetary-taste analysis is therefore to separate them.

The broader lesson is equally important. A resilient monetary system need not require the disappearance of any particular currency. What matters is whether economic agents possess meaningful choice, interoperability, substitution possibilities, and institutional capacity.

In this sense, the ultimate objective is not to determine which currency the world *ought* to prefer.

It is to understand why the world chooses what it chooses.

$$\boxed{\text{Monetary Taste} = \text{Preference} + \text{History} + \text{Institutions} + \text{Networks} + \text{Strategy.}} \quad (102)$$

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Glossary

Currency Exposure

The observed economic or financial position of an agent with respect to a currency.

Currency Preference

A latent ordering over currencies or monetary instruments conditional on comparable choice conditions.

Monetary Taste

A multidimensional representation of preferences over monetary characteristics, including liquidity, stability, convertibility, return, network access, institutional credibility, and strategic utility.

Network Externality

An effect whereby the usefulness of an economic instrument changes with the number or importance of other users.

Revealed Preference

Inference about preferences from observed choices, conditional on the relevant choice set and constraints.

Monetary Dependence

A structural condition in which economic activity is substantially exposed to a monetary instrument or monetary network.

Path Dependence

A process in which historical choices influence subsequent feasible choices or equilibrium outcomes.

Liquidity

The ability to transact an asset rapidly with relatively low transaction cost and price impact.

Numeraire

The unit in which economic values are expressed.

Reserve Currency

A currency held by monetary authorities as part of official international reserves.

Systemic Concentration

The concentration of economically important exposures in a small number of instruments or institutions.

Monetary Pluralism

An international monetary architecture in which multiple monetary instruments coexist.

Latent Variable

A variable that is not directly observed but is inferred from observed indicators.

Revealed Monetary Preference

An empirically inferred preference structure derived from observed monetary choices after accounting for relevant constraints.

Network Dominance

A condition in which a monetary instrument has extensive use partly because extensive use itself increases its usefulness.

Monetary Resilience

The capacity of a monetary system to continue functioning under financial, economic, technological, or geopolitical shocks.

The End